16. A complex in which the oxidation number of the metal is zero is

a) $K_4[Fe(CN)_6]$ b) $[Fe(CN)_3(NH_3)_3]$

c) $[Fe(CO)_5]$ d) both (b) and (c)

17. Formula of tris(ethane-1,2-diamine)iron(II)phosphate

a) $[Fe(CH_3\text{-}CH(NH_2)_2)_3](PO_4)_3$ b) $[Fe(H_2N\text{-}CH_2\text{-}CH_2\text{-}NH_2)_3](PO_4)$

c) $[Fe(H_2N\text{-}CH_2\text{-}CH_2\text{-}NH_2)_3](PO_4)_2$ d) $[Fe(H_2N\text{-}CH_2\text{-}CH_2\text{-}NH_2)_3]_3(PO_4)_2$

18. Which of the following is paramagnetic in nature?

a) $[Zn(NH_3)_4]^{2+}$ b) $[Co(NH_3)_6]^{3+}$

c) $[Ni(H_2O)_6]^{2+}$ d) $[Ni(CN)_4]^{2-}$

19. Fac-mer isomerism is shown by

a) $[Co(en)_3]^{3+}$ b) $[Co(NH_3)_4(Cl)_2]^{+}$

c) $[Co(NH_3)_3(Cl)_3]$ d) $[Co(NH_3)_5Cl]SO_4$

20. Choose the correct statement.

a) Square planar complexes are more stable than octahedral complexes

b) The spin only magnetic moment of $[Cu(Cl)_4]^{2-}$ is 1.732 BM and it has square planar structure.

c) Crystal field splitting energy (Δ_0) of $[FeF_6]^{4-}$ is higher than the (Δ_0) of $[Fe(CN)_6]^{4-}$

d) crystal field stabilization energy of $[V(H_2O)_6]^{2+}$ is higher than the crystal field stabilization of $[Ti(H_2O)_6]^{2+}$

Answer the following questions:

1. Write the IUPAC names for the following complexes.

i) $Na_2[Ni(EDTA)]$

ii) $[Ag(CN)_2]^{-}$

iii) $[Co(en)_3]_2(SO_4)_3$

iv) $[Co(ONO)(NH_3)_5]^{2+}$

v) $[Pt(NH_3)_2Cl(NO_2)]$

XII U5 Coordination jagan.indd 171 04-01-2021 13:24:37